

JUNMING REN

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EDUCATION

Fudan University

Expected June 2027

School of Data Science | Bachelor of Data Science and Big Data Technology

- **Grade:** GPA: 3.62/4.0, TOEFL: 5.0
- **Honors:** Third Prize Scholarship of Fudan University, Honorable Prize in COMAP 2026

University of Edinburgh

Sept. 2025 – Dec. 2025

School of Informatics | Exchange Student

ACADEMIC RESEARCH

Ethereum Arbitrage Transaction Detection

Feb. 2025 – Sep. 2026

Submitted to the *IEEE PacificVis 2027 TVCG Track*; First Author and Lead Researcher

Conducted at the *FDU-VIS Lab*; Funded by *FDUROP*

- Reconstructed dynamic control-flow graphs along actual transaction execution paths and applied structure-aware folding to reduce graph complexity while preserving key execution semantics.
- Recovered ERC-20 and ETH transfers directly from opcode-level EVM execution traces without relying on contract source code, ABIs, or event logs.
- Developed TraceWeaver, an interactive visual analytics system that links token-flow graphs, call trees, and control-flow graphs, enabling arbitrage fund flows to be traced to specific contract calls and opcodes.

Minimizing Redundant Search Cost for Continuous Subgraph Matching

Feb. 2026 – May 2026

- Implemented query graph preprocessing to identify three types of substructures reused in the search process.
- Visualized the algorithmic workflow and experimental results and contributed to manuscript writing.

WORK EXPERIENCE

Shanghai Jiayicui Private Fund Management Co., Ltd.

May 2026 – Jul. 2026

Quantitative Research Intern

- Built a cross-sectional convertible-bond return ranking model with multi-factor encoding, a multi-relational GNN, and an MLP head; evaluated the incremental value of different graph relations.
- Achieved a test-set RankIC of 0.095 through regularization, hyperparameter optimization, and ensembling models using different graph relations.
- Optimized factor processing, matrix computation, and graph aggregation with Polars and GPU acceleration, reducing data-processing and I/O overhead.

SELECTED PROJECTS

- ERC-8004 Ecosystem Visualization | NFT Music: Wikipedia Article
- Vicky NFT Auction Contract | ThreeBody-ZHAOGE: Multi-Agent Dialogue Game

TEACHING & CAMPUS SERVICE

Teaching Assistant, Crypto Finance (English-Taught Graduate Course)

Sep. 2026 – Dec. 2026

- Designed a hands-on Sepolia-based midterm assignment and guided students through wallet setup, test-token acquisition, NFT minting, and marketplace sales; led in-class Q&A and assisted with final-paper reviews.

Athletics Coordinator, School of Data Science

Sep. 2024 – Present

- Coordinated the School's registration and participation in university-wide athletic events.

TECHNICAL SKILLS

Programming: Python (PyTorch, Polars), SQL, Solidity, C | **Tools:** Git, L^AT_EX